BCS-303	DIGITAL ELECTRONICS	L T P 2 1 0	3 credits

Course Objectives:

CO1	Understand the concepts of various components.
CO2	Understand concepts that underpin the disciplines of analog and digital electronic logic circuits.
CO3	Understand various Number systems and Boolean algebra, the Boolean expression using Boolean algebra and design it using logic gates.
CO4	Understand Design and implementation of combinational circuits
CO5	Understand Design and develop sequential circuits.

MODULE-I

Digital system and binary numbers: Signed binary numbers, binary codes, cyclic codes, error detecting and correcting codes, hamming codes. Floating point representation Gate-level minimization: The map method up to five variable, don't care conditions, POS simplification, NAND and NOR implementation, Quine Mc-Clusky method (Tabular method).

MODULE-II

Combinational Logic: Combinational circuits, analysis procedure, design procedure, binary adder-subtractor, decimal adder, binary multiplier, magnitude comparator, decoders, encoders, multiplexers
Synchronous Sequential logic: Sequential circuits, storage elements: latches, flip flops, analysis of clocked sequential circuits, state reduction and assignments, design procedure. Registers and counters: Shift registers, ripple counter, synchronous counter, other counters.

MODULE-III

Memory and programmable logic: RAM, ROM, PLA, PAL.
Design at the register transfer level: ASMs, design example, design with multiplexers.
Asynchronous sequential logic: Analysis procedure, circuit with latches, design procedure, reduction of state and flow table, race free state assignment, hazards.

Text Book:

1. M. Morris Mano and M. D. Ciletti, "Digital Design", 4th Edition, Pearson Education

Reference Books :

1. Introduction to Digital Logic Design, JP Hayes, PHI.

2. The Art of Digital Design: An Introduction to Top-Down Design, Franklin P. Prosser, PHI.

COURSE OUTCOMES:

CO1	Understand the concepts of various components to design stable analog circuits.
CO2	Represent numbers and perform arithmetic operations.
CO3	Minimize the Boolean expression using Boolean algebra and design it using logic gates.
CO4	Analyze and design combinational circuit.
CO5	Design and develop sequential circuits.